The 1st Global University Student Life Science Challenge Project Abstract

AI-Assisted Discovery of Novel Small-Molecule Modulators of GPR68 for Cartilage Protection in Osteoarthritis

Competition Region: East China Region
Competition Group: Undergraduate Group
Track: 3 Virtual Screening of Small Molecule Drugs Using Ion Channels or GPCR Drug

Targets

Project Summary:

Osteoarthritis (OA) involves cartilage degeneration, inflammation and extracellular matrix degradation within an increasingly acidic local microenvironment. GPR68, a proton-sensing G protein-coupled receptor, offers a potential route to targeting this microenvironment. This project aims to identify small-molecule GPR68 modulators with cartilage-protective potential through AI-assisted structural screening followed by receptor-level and disease-relevant experimental validation. A commercial library of 10,240 compounds was successfully processed using RDKit, with no failed structures or duplicate canonical SMILES detected. Molecular descriptors were calculated to characterise the library. A human GPR68 receptor model was prepared from the experimental structure PDB 8Z68. Machine-learning-based pocket prediction using P2Rank identified Pocket 1 and Pocket 2 as the leading candidate binding regions, with scores of 7.06 and 6.87, respectively. The GNINA implementation, incorporating pretrained convolutional neural network scoring, was technically verified using a standard redocking benchmark, yielding a best-pose RMSD of approximately 0.59 Å. This result confirms workflow operation under the benchmark conditions; it does not establish GPR68-specific docking validity. Benchmarking with known GPR68 ligands has not yet been completed and will compare the two candidate pockets under identical docking conditions. Following pocket assessment and library screening, 6–10 chemically diverse candidates will be prioritised for GPR68 functional assays, pH-dependent response measurements and OA-related chondrocyte experiments. GPR68-dependent experiments will test whether any observed cartilage-protective effects are mediated by the receptor. These initial results provide a computational foundation for an integrated discovery strategy linking the OA acidic microenvironment, AI-assisted screening and experimental evaluation of potential GPR68 modulators.

Keywords: osteoarthritis; GPR68; AI-assisted drug discovery; virtual screening; cartilage protection